$\sqrt{2}$

27 B *F = Magnetic force on YZ + Magnetic force on XY*

$F = F_{XY} + F_{YZ}$

As the component of L_{XY} which is perpendicular to the magnetic field is the same as L,

$F_{XY} = F_{YZ} = \frac{F}{2}$

$F_{XY} = \frac{F}{2} = BIL_{XY}\cos\theta$ (as $L = L_{XY}\cos\theta$)

$L_{XY} = \frac{F}{2BI\cos\theta}$